

General Questions

1. **What is the purpose of the IPL Score Predictor project?**

- Answer: The purpose of the project is to predict cricket scores in IPL matches based on historical data and various match conditions, such as overs, wickets, and run rate.

2. **What data sources did you use for this project?**

- Answer: The project uses historical IPL match data, which may include features such as runs scored, overs bowled, wickets lost, and team-specific statistics.

3. **What tools or libraries did you use for this project?**

- Answer: The project is implemented using Jupyter Notebook, with Python libraries like pandas, NumPy, scikit-learn, and matplotlib for data processing, modeling, and visualization.

Data Preprocessing

4. **How did you handle missing or inconsistent data in your dataset?**

- Answer: Missing data was handled by techniques like imputation (filling missing values with the mean/median/mode) or by dropping rows/columns if the missing data was significant.

5. **What features were most important for predicting the IPL scores?**

- Answer: Key features included overs, current runs, wickets lost, run rate, and team-specific performance metrics.

Modeling

6. **What machine learning models did you use for score prediction?**

- Answer: Regression models such as Linear Regression, Decision Tree Regression, or Random Forest Regression were used to predict the target variable (final score).

7. **How did you evaluate the performance of your model?**

- Answer: Metrics like Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared were used to evaluate the performance.

8. **Did you perform any feature engineering? If yes, give an example.**

- Answer: Yes, features like the current run rate (runs/overs) were derived to help the model better understand the scoring pace.

Challenges and Improvements

9. **What challenges did you face in this project?**

- Answer: Challenges included handling imbalanced data, selecting meaningful features, and avoiding overfitting during model training.

10. **If given more time, how would you improve this project?**

- Answer: Improvements could include integrating real-time match data, using ensemble models for better predictions, and creating a user-friendly dashboard for visualization.

Data Science Concepts

11. **What is the difference between overfitting and underfitting, and how did you address them?**

- Answer: Overfitting occurs when the model performs well on training data but poorly on test data, while underfitting occurs when the model fails to capture the underlying patterns. Techniques like cross-validation, regularization, and hyperparameter tuning were used to address these issues.

12. **Explain how you split your dataset for training and testing.**

- Answer: The dataset was split into training and testing sets using an 80-20 or 70-30 ratio to ensure the model was evaluated on unseen data.

Project Deployment

13. ****Did you deploy the model? If yes, how?****

- Answer: The model can be deployed using Flask or FastAPI to serve predictions via a web interface, or integrated into a cloud platform like AWS or Google Cloud.

14. ****How would you ensure the model remains accurate over time?****

- Answer: Regular updates with new data and periodic retraining of the model would ensure accuracy over time.

**Scenario-Based Questions**

15. ****How would your model handle a scenario where a team has lost 8 wickets but still has 5 overs left?****

- Answer: The model would consider the current run rate and wickets lost to predict a lower final score due to a higher likelihood of dismissal.

16. ****What would you do if your model predictions were consistently off by 10-15 runs?****

- Answer: Investigate feature importance, re-tune hyperparameters, or try a more complex model to improve accuracy.

**Visualization and Insights**

17. ****What visualizations did you create to understand the data better?****

- Answer: Visualizations like scatter plots for runs vs. overs, bar charts for team performance, and heatmaps for correlations were created.

18. ****Can you provide an example of a key insight from your data analysis?****

- Answer: Teams with a higher run rate during the middle overs (7-15) tend to achieve higher final scores, highlighting the importance of maintaining a steady scoring pace.